

LOS + LMS Prototype - High-Level Design

LMS Prototype

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LOS + LMS Prototype — High-Level Design (HLD)

Date: 2026-08-05 **Companion docs:** [Spec](#) · [LLD](#) · [Plan](#)

1. Purpose

Deliver a deployable prototype demonstrating the end-to-end personal-loan lifecycle: application → underwriting decision → disbursement → amortization schedule → installment payment → balance decrement. The prototype must be demoable against a single hosted URL in under two minutes, with seed data that exercises both origination and mid-repayment flows without manual setup.

2. Scope

In scope (this prototype):

- One loan product (fixed-rate personal loan)
- Two user roles: `applicant` , `admin`
- Email + password authentication (JWT in `httpOnly` cookie)
- Rule-based underwriting (FOIR + income-multiple) with admin override
- Amortization schedule generation, installment payment, outstanding balance tracking
- Seeded admin and two applicants (one fresh, one mid-flight) for walkthrough

Out of scope (deferred): real KYC, credit bureau, document upload, email/SMS, late fees, prepayment, multi-product, multi-tenancy, audit log, PDF statements, real payments. See spec §16.

3. System architecture

flowchart LR

```
Browser[Browser<br>React 19 + shadcn] -->|HTTPS| Render[Render Node service<br>Fastify]
Render -->|serves| Static[Built React SPA<br>apps/web/dist]
Render -->|/api/* JSON| API[Fastify routes<br>Zod-validated]
API -->|Drizzle ORM| SQLite[(SQLite file<br>/data/lms.db)]
API -->|httpOnly cookie<br>lms_session| Browser
Admin[Admin browser tab] -->|same URL, /admin/*| Render
```

Topology: one Node process, one port (10000), one URL. Fastify serves `/api/*` from the same process that serves the built React bundle for every other path. There is no second service, no CORS, no reverse proxy. SQLite is a single file on a Render persistent disk.

Process responsibilities:

- **Fastify process** — HTTP server, JSON API, JWT verification, Zod validation, route handlers, Drizzle queries, domain logic (amortization, underwriting), static-file serving for the SPA.
- **SQLite file** — single source of truth. No replicas, no backups beyond Render's disk snapshots.
- **React SPA** — every screen, including `/admin/*`, runs in the same client. Role-based route guards determine what the user sees.

4. Technology stack

Layer	Choice	Why
Frontend	React 19, Vite, TypeScript, Tailwind, shadcn	Modern, typesafe, single component model
Routing	React Router 6	Mature, shadcn examples assume it
Server state	TanStack Query	Cache + invalidation without Redux/Zustand
Forms	react-hook-form + zodResolver	Uncontrolled, works with shared Zod schemas
Backend	Fastify, TypeScript, Node 20 LTS	Fast, typesafe, plugin model
Validation	Zod	One schema, two consumers (API + web)
ORM	Drizzle	Typesafe queries, plain SQL, easy migrations
Database	SQLite (better-sqlite3)	Zero-config, file-based, fine for one writer
Auth	@fastify/jwt + @fastify/cookie + bcrypt	httpOnly cookie, no CSRF for our routes
Workspace	pnpm workspaces	Single lockfile, no Turborepo overhead
Deploy	Render single Node service	Free tier, persistent disk, single env

5. Data flow

5.1 Apply → Underwrite → Disburse

```
sequenceDiagram
    actor A as Applicant
    actor AD as Admin
    participant W as Web (React)
    participant API as Fastify
    participant DB as SQLite

    A->>W: fill /apply multi-step form
    W->>API: POST /api/applications
    API->>API: validate (Zod) + compute EMI
    API->>API: run underwriting rule
    alt rule rejects
        API->>DB: INSERT status='rejected'
        API-->>W: 201 { application, decision_reason }
    else rule marks pending
        API->>DB: INSERT status='pending'
        API-->>W: 201 { application }
    end

    AD->>W: open /admin queue
    W->>API: GET /api/admin/applications?status=pending
    API-->>W: applications[]
    AD->>W: review, click Approve
    W->>API: POST /api/admin/applications/:id/decision { approve }
    API->>DB: UPDATE status='approved', decided_by, decided_at
    AD->>W: click Disburse
    W->>API: POST /api/admin/applications/:id/disburse
    API->>DB: BEGIN
    API->>DB: INSERT loans, status='active'
    API->>API: generate amortization schedule
    API->>DB: INSERT installments (1..N)
    API->>DB: UPDATE applications status='disbursed'
    API->>DB: COMMIT
    API-->>W: 200 { loan, installments }
```

5.2 Pay EMI

```
sequenceDiagram
    actor A as Applicant
    participant W as Web
    participant API as Fastify
    participant DB as SQLite

    A->>W: open /loans/:id
    W->>API: GET /api/loans/:id
    API-->>W: { loan, installments }
    A->>W: click "Pay EMI" on next unpaid row
```

```

W->>API: POST /api/loans/:id/installments/:n/pay
API->>DB: BEGIN
API->>DB: SELECT installment WHERE sequence = :n AND paid_at IS NULL
alt not found
  API->>DB: ROLLBACK
  API-->>W: 409 { error: 'NoInstallmentFound' }
else found
  API->>DB: UPDATE installment SET paid_amount, paid_at
  API->>DB: UPDATE loans SET outstanding_cents -= amount
  API->>DB: if last installment: UPDATE loans.status='closed'
  API->>DB: COMMIT
  API-->>W: 200 { installment, loan }
end
W->>W: invalidate ['loans', id] query

```

6. External integrations

None. This is a self-contained prototype. There are no third-party services — no payment processor, no email provider, no credit bureau, no KYC vendor, no analytics. The brief allows for a "live, interactive environment"; this design delivers that entirely on Render + a single SQLite file.

Future work (out of scope) would add: Stripe (payments), SendGrid (email), Plaid (bank linking), Onfido/Persona (KYC), Experian/Equifax (credit). The HLD does not plan for these.

7. Security

Concern	Approach
Password storage	bcrypt with cost factor 10
Auth tokens	JWT in httpOnly + SameSite=Lax cookie (<code>lms_session</code>)
CSRF	Not needed: cookie is SameSite=Lax, no state-changing GETs
XSS	React's default escaping + shadcn primitives; no <code>dangerouslySetInnerHTML</code>
SQL injection	Drizzle parameterizes all queries
Secrets	<code>JWT_SECRET</code> and <code>DATABASE_URL</code> from env, never in repo
CORS	Not needed (same origin)
Rate limiting	Not implemented (out of scope; Render edge handles DDoS)

Concern	Approach
Audit log	Not implemented (out of scope)

8. Deployment topology

flowchart TB

```

subgraph Render["Render (us-east-1, free tier)"]
  Disk[Persistent disk<br/>/data, 1 GB]
  Web[Node web service<br/>Fastify + built SPA]
  Web --- Disk
end
Browser[End user] -->|HTTPS| Web
GitHub[GitHub repo<br/>main branch] -->|auto-deploy on push| Web

```

- **Build:** `pnpm install --frozen-lockfile && pnpm build`. Produces `apps/web/dist` and `apps/api/dist`.
- **Start:** `pnpm start` runs `node apps/api/dist/server.js`. Before serving, the start script runs `migrate` then `seed` (idempotent upsert), so a cold start always ends with a usable DB.
- **Persistent disk:** mounted at `/data`, `DATABASE_URL=/data/lms.db`. Render's free tier provides 1 GB; the seed DB is < 100 KB.
- **Auto-deploy:** Render watches the GitHub repo's `main` branch. Push → build → deploy.
- **Env vars (set in Render dashboard, mirrored in `.env.example`):**
 - `DATABASE_URL` — path to the SQLite file
 - `JWT_SECRET` — random 64-char hex
 - `NODE_ENV=production`
 - `PORT=10000`
- **No CI.** Render handles build + deploy. Tests run locally before push.

9. Observability

Out of scope. The free Render service exposes request logs via the dashboard; no structured logging, no metrics, no tracing. The walkthrough video and `pnpm test` are the prototype's "is it working?" signals.

10. Future work (not built)

Anything in the spec's "out of scope" list is fair game for a v2: real KYC, real payments, document upload, multi-product, audit log, PDF statements, email notifications, late-fee calculation, prepayment, mobile-first polish.

End of HLD. For schemas, signatures, and per-module specs see the [LLD](#). For phased implementation see the [Plan](#).